PROJECT-BASED LEARNING

PHASE-I EVALUATION

VERI NEWS

Team ID: DSCPP-III-2026-T224

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

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Team & Mentor Details

Team Information

Team ID: DSCPP-III-T224
Semester: 3rd
Section: D, J, ML2
Domain: Data Structures + OOP / C++

Team Members

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Mentor

Mentor Name: Mr. Kamal Kumar Gola

Interactions completed: ____________

Problem Statement & Motivation

Problem Statement

Online information can spread fast, making it hard for a user to determine what news is reliable and what is simply fake or suspicious. Manual verification can be tedious and error-prone.

Motivation

- Rapid spread of unverified information
- Difficulty of checking every claim manually
- Need for a simple preliminary verification tool
- Need to compare claims with stored evidence

Target Users / Stakeholders

- General internet users
- Students and researchers
- News readers
- Content moderators

Objectives

Project Objectives

1. Develop a solution for preliminary news verification.
2. Store and manage verified and suspicious news records.
3. Use DSA for efficient searching, indexing and matching.
4. Apply C++ OOP for modular application design.
5. Generate an explainable credibility / matching score.
6. Evaluate the system with predefined test cases.

Key Objective: Provide users with a quick, explainable preliminary assessment of a news claim based on stored records and matching evidence.

Proposed Solution

System Idea

The system checks a news claim by extracting keywords, finding related stored records, and giving a preliminary result.

Main Features

- News verification
- Keyword search
- Similar-news detection
- Source checking
- Verification report

Processing

1. Extract keywords $\rightarrow$ Hash for lookup
2. Find related news $\rightarrow$ Search & rank
3. Calculate matching score

Input

- Headline or news text
- Existing news records

Result

- Likely Verified
- Needs Verification
- Suspicious

Stored: News records | Sources | Status | Search history

Integration of PBL Course Concepts

PBL Integration

DSA provides efficient storage and searching, while C++ OOP organizes the system into reusable classes and modules.

Concept	Project Use
DSA in C	Linked List, Hashing, BST, Searching, Sorting
OOP with C++	Classes, Objects, Encapsulation, Inheritance, Polymorphism
File Handling	News records, user activities, and verification logs
Database	News data and verified source records

DSA Module

- **Linked List:** News records
- **Hash Table:** Fast keyword lookup
- **BST / Queue:** News indexing & requests
- **Sorting:** Match scoring and ranking

C++ Classes

- **User** & **News**
- **NewsDatabase** & **NewsAnalyzer**
- **VerificationEngine**
- **Report**

System Architecture / Workflow

Input Claim / Text → Processing Layer → Core Search Logic → Database Layer → Verification Result

Data / Control Flow

- User inputs claim or article text for verification.
- Text is cleaned and keywords are extracted.
- Hashing and BST search find related stored records.
- Relevance scoring generates a final classification.

Major System Components

- **Input Module:** Receives raw headline or claim.
- **Analyzer Module:** Tokenization and stopword removal.
- **Matching Engine:** Compares keywords with database.

Processing and Storage Pipeline

- **Input Layer:** Claim text entered by the user.
- **Processing Layer:** Validates structure and extracts tokens.
- **Core Logic:** Hash lookup, searching, and credibility score.
- **Database Layer:** Stores known news and trusted sources.
- **Output Layer:** Credibility score and summary status.

Technology Stack & Methodology

Technology Stack

- **C & C++**: DSA in C and OOP structuring in C++.
- **DSA**: Hashing, BST, searching, and sorting.
- **File Handling**: Stores local database and logs.
- **VS Code / Code::Blocks**: IDE and build environment.
- **Git & GitHub**: Version control and team tracking.

Development Methodology

1. **Requirements:** Define parameters, inputs, and outputs.
2. **Design:** Model classes, hash functions, and trees.
3. **Implementation:** Build modular C/C++ components.
4. **Testing:** Verify keyword search with edge cases.
5. **Evaluation:** Measure score accuracy and latency.

Purpose

Build a modular, searchable, and explainable news verification prototype.

Pipeline: Requirements ⟶ Design ⟶ Implementation ⟶ Testing ⟶ Evaluation

Initial Progress & Team Contribution

Progress Achieved

- Problem statement identified
- Background study completed
- System requirements drafted
- Architecture and data flow designed
- Core DSA concepts mapped
- Database file schemas established
- Prototype implementation scheduled

Individual Contribution

Member	Role	Contribution
Shambhavi Srivastava	Team Lead	Project planning, requirements, and system design coordination
Abhinav Pan-war	Developer	C++ OOP classes, indexing, and news data management
Akshat Sharma	Testing / Docs	Test suite design, verification metrics, and documentation

Roadmap, Expected Outcomes & References

Roadmap

1. **Phase-I (Analysis & Design):** Define news schemas, select DSA/OOP structures, and design verification pipeline.
2. **Phase-II (Development):** Implement modules, hash indexing, matching engine, and file-based data store.
3. **Phase-III (Testing & Finalization):** Run structured news test cases, optimize search times, and deliver working tool.

Expected Outcomes

- Working TruthCheck preliminary verification engine
- Sub-second keyword lookups via hashing
- Transparent, rule-based credibility indicators
- Foundation for future automated NLP ingestion

References

1. Data Structures & Algorithms – Course Notes
2. Object-Oriented Programming with C++ – Reference Guide
3. ISO/IEC C++ Standard Documentation
4. GitHub Repository for Team Version Tracking

Thank You

Questions & Suggestions